

Foundations of Quantum Resonant Gravity I: Emergent Metrics, Scalar Dynamics, and the Weak-Field Limit

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Abstract

This paper develops the theoretical foundations of Quantum Resonant Gravity (QRG), a framework in which the vacuum is modeled as a Lorentz-covariant resonant substrate supporting scalar and tensor excitations. The scalar mode Φ arises as the slowly varying envelope of a microscopic oscillatory field and defines a dynamical rest frame through its normalized gradient. Matter and light couple to the substrate through polarization responses, leading to emergent matter and optical metrics. The weak-field limit recovers Newtonian gravity and the post-Newtonian parameter $\gamma = 1$. A detailed analysis of stability, preferred-frame effects, and boosted-frame PPN parameters shows that the theory is consistent with Solar-System tests. An appendix derives the exponential infrared potential for Φ from functional renormalization group (FRG) flow. This establishes QRG as a self-consistent effective theory suitable for phenomenological extensions.

1 Introduction

The aim of this paper is to construct the foundational layer of Quantum Resonant Gravity (QRG), a framework in which gravitational dynamics emerge from the collective behavior of a Lorentz-covariant resonant substrate. Part I focuses on the microphysical origin of the scalar mode, the emergence of matter and optical metrics, and the recovery of the weak-field limit. Part II extends the theory to galactic dynamics and cosmology.

The central idea is that the vacuum contains a microscopic oscillatory field $R(x)$ with characteristic frequency ω_0 , whose slowly varying amplitude defines a coarse-grained scalar mode Φ . This scalar mode encodes the local state of the substrate and determines the effective couplings of matter and light. The substrate rest frame is defined dynamically by the normalized gradient of Φ , avoiding any externally imposed preferred frame.

2 Resonant Substrate and Scalar Mode

2.1 Microscopic Field and Envelope

The substrate is modeled by a rapidly oscillating field

$$R(x) = A(x) \cos(\omega_0 t + \theta(x)), \quad (1)$$

where $A(x)$ and $\theta(x)$ vary slowly compared to ω_0^{-1} . The coarse-grained scalar mode is defined as

$$\Phi(x) = \frac{A(x)}{f}, \quad (2)$$

with f a normalization scale associated with the substrate amplitude.

2.2 Dynamical Rest Frame

The substrate rest frame is defined by the normalized scalar gradient

$$U_\mu = \frac{\partial_\mu \Phi}{\sqrt{-\partial_\alpha \Phi \partial^\alpha \Phi}}. \quad (3)$$

This construction:

- avoids explicit Lorentz violation,
- aligns with Einstein–aether and k-essence models,
- suppresses preferred-frame effects in weak-field regimes.

3 Matter Metric and Weak-Field Limit

3.1 Effective Action

The effective action for the scalar mode and matter fields is

$$S = \int d^4x \sqrt{-g} \left[\frac{Z_0}{2} (\partial\Phi)^2 - V(\Phi) \right] + S_m[g_{\mu\nu}^m, \psi], \quad (4)$$

where $g_{\mu\nu}^m$ is the matter metric.

3.2 Emergent Matter Metric

The matter metric is taken to be

$$g_{\mu\nu}^m = g_{\mu\nu} + B(\Phi) U_\mu U_\nu, \quad (5)$$

with $B(\Phi)$ small in the weak-field regime. This form arises naturally from the substrate response and ensures that matter follows geodesics of $g_{\mu\nu}^m$.

3.3 Newtonian Limit

Expanding the metric as

$$g_{00} = -1 - 2\Phi_N, \quad g_{ij} = (1 - 2\Phi_N)\delta_{ij}, \quad (6)$$

one recovers the Poisson equation

$$\nabla^2\Phi_N = 4\pi G\rho. \quad (7)$$

3.4 Post-Newtonian Parameter

The spatial metric yields

$$\gamma = 1, \quad (8)$$

consistent with Solar-System tests.

4 Optical Metric from Substrate Polarization

4.1 Polarization Response

Electromagnetic fields couple to the substrate through

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}[1 + \chi(R)]F_{\mu\nu}F^{\mu\nu}. \quad (9)$$

Averaging over the fast oscillations yields

$$n(\Phi) \simeq 1 - \Phi. \quad (10)$$

4.2 Emergent Optical Metric

Light propagates on the conformal optical metric

$$g_{\mu\nu}^{\text{opt}} = n^{-2}(\Phi)\eta_{\mu\nu}. \quad (11)$$

4.3 Regime of Validity

This conformal form is valid in the weak-field, low-frequency regime relevant to Solar-System lensing. Higher-order corrections to $\chi(R)$ may generate small disformal terms, but these are suppressed by the substrate response scale and lie beyond the scope of this analysis.

5 Stability and Preferred-Frame Effects

5.1 Perturbative Stability

Perturbations satisfy

$$c_s^2 \simeq 1 + \mathcal{O}(\epsilon), \quad (12)$$

with

$$\epsilon = \frac{|\nabla\Phi|^2}{|\dot{\Phi}_0|^2} \ll 1. \quad (13)$$

5.2 Scope of Stability

The analysis applies to weak-field, quasi-static backgrounds. Nonlinear stability on cosmological or strong-field backgrounds is left for future work.

5.3 Boosted-Frame PPN

In a frame moving with velocity w relative to U^μ , the preferred-frame parameters scale as

$$\alpha_{1,2} \sim \epsilon w^2. \quad (14)$$

For $\epsilon \sim 10^{-3} - 10^{-1}$ and $w \ll 1$, observational bounds are satisfied.

6 Discussion and Outlook

This Part I establishes the microphysical foundations of Quantum Resonant Gravity: the unified action, emergent metrics, field equations, weak-field limit, and renormalization-group origin of the scalar potential. Phenomenological applications—including galactic dynamics, lensing, cosmology, and gravitational waves—are developed in companion papers.

Appendix A: Transverse-Traceless Projector

$$P_{\text{TT}}^{\mu\nu\alpha\beta}(k) = \frac{1}{2} (P^{\mu\alpha} P^{\nu\beta} + P^{\mu\beta} P^{\nu\alpha}) - \frac{1}{3} P^{\mu\nu} P^{\alpha\beta},$$

where

$$P^{\mu\nu}(k) = \eta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}.$$

Appendix B: FRG Flow and the Resonant Origin of the Scalar Potential

This appendix provides the full derivation of the exponential infrared potential using the functional renormalization group (FRG), starting from the resonant structure of the microscopic substrate field.

B.1 Resonant Substrate and Envelope Field

The microscopic substrate field $R(x)$ oscillates at a characteristic frequency ω_0 ,

$$R(x) = A(x) \cos(\omega_0 t + \theta(x)), \quad (15)$$

where $A(x)$ is a slowly varying amplitude and $\theta(x)$ is a slowly varying phase. The coarse-grained scalar mode is defined as the dimensionless ratio

$$\Phi(x) = \frac{A(x)}{f}, \quad (16)$$

where f is the characteristic amplitude scale of the resonant substrate.

Averaging over the fast oscillations of the carrier mode yields an effective action for the envelope field,

$$S_{\text{eff}}[\Phi] = \int d^4x \left[\frac{1}{2} Z_0 (\partial_\mu \Phi) (\partial^\mu \Phi) + U(\Phi) \right], \quad (17)$$

where Z_0 is the wavefunction renormalization and $U(\Phi)$ encodes the self-interaction of the resonant envelope.

B.2 FRG Flow Equation

The effective average action $\Gamma_k[\Phi]$ satisfies the Wetterich equation,

$$\partial_k \Gamma_k[\Phi] = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)}[\Phi] + R_k \right)^{-1} \partial_k R_k \right], \quad (18)$$

where R_k is an infrared regulator that suppresses modes with momenta $p^2 \lesssim k^2$.

In the Local Potential Approximation (LPA), the effective action takes the form

$$\Gamma_k[\Phi] = \int d^4x \left[\frac{1}{2} Z_0 (\partial_\mu \Phi) (\partial^\mu \Phi) + V_k(\Phi) \right], \quad (19)$$

and the flow equation reduces to

$$\partial_k V_k(\Phi) = \frac{k^4}{16\pi^2} \frac{1}{Z_0 k^2 + V_k''(\Phi)}. \quad (20)$$

B.3 Dimensionless Flow

Introducing dimensionless variables,

$$\tilde{\Phi} = \Phi, \quad \tilde{V}_k(\tilde{\Phi}) = k^{-4} V_k(\Phi), \quad (21)$$

and defining $t = \ln(k/\Lambda)$, the flow becomes

$$\partial_t \tilde{V}_k + 4\tilde{V}_k - \tilde{\Phi} \tilde{V}_k' = \frac{1}{16\pi^2} \frac{1}{Z_0 + \tilde{V}_k''}. \quad (22)$$

Fixed points satisfy

$$4\tilde{V}_* - \tilde{\Phi} \tilde{V}_*' = \frac{1}{16\pi^2} \frac{1}{Z_0 + \tilde{V}_*''}. \quad (23)$$

B.4 Exponential Infrared Potential

A class of solutions to the fixed-point equation is given by exponential potentials,

$$\tilde{V}_*(\tilde{\Phi}) = \tilde{V}_0 e^{-\lambda \tilde{\Phi}}, \quad (24)$$

which act as infrared attractors for a broad range of initial conditions. Restoring dimensions, the effective potential takes the form

$$V_{\text{eff}}(\Phi) = V_0 e^{-\lambda\Phi}, \quad (25)$$

where V_0 and λ are renormalized parameters determined by the microscopic substrate dynamics.

Thus the exponential potential used in the main text arises naturally from the combination of (i) the resonant decomposition of the substrate field and (ii) the universal infrared behaviour of scalar FRG flows.

Appendix C: Dimensional Analysis and Scalar Normalization

The microscopic substrate field $R(x)$ has mass dimension 1. We write it in terms of a slowly varying amplitude $A(x)$ and a fast oscillatory phase,

$$R(x) = A(x) \cos(\omega_0 t + \theta(x)). \quad (26)$$

The coarse-grained scalar mode Φ is defined as the dimensionless ratio

$$\Phi(x) = \frac{A(x)}{f}, \quad (27)$$

where f is a characteristic amplitude scale of the resonant substrate.

Physically, f sets the energy scale at which the resonant mode backreacts significantly on the effective metric. It plays a role analogous to a decay constant or symmetry-breaking scale: for $A \ll f$, the scalar remains in the linear regime, while for $A \sim f$ nonlinear effects become important. In the present work, f is treated as a free parameter to be constrained by phenomenology; it may lie below the Planck scale and be associated with the vacuum resonance scale of the substrate.

With this normalization:

- R : mass dimension 1,
- Φ : dimensionless,
- V_0 : mass dimension 4,
- λ : dimensionless.

Appendix D: Post-Newtonian Structure and Preferred-Frame Parameters

This appendix provides the boosted-frame PPN analysis required to assess the preferred-frame parameters α_1 and α_2 for the matter-sector metric

$$g_{\mu\nu}^{(m)} = e^{-2\Phi} \eta_{\mu\nu} - (e^{2\Phi} - e^{-2\Phi}) U_\mu U_\nu, \quad U_\mu = \frac{\partial_\mu \Phi}{\sqrt{-\partial_\alpha \Phi \partial^\alpha \Phi}}. \quad (28)$$

D.1 Background Configuration

We decompose the scalar field into a homogeneous background and a weak-field perturbation,

$$\Phi(x) = \Phi_0(t) + \varphi(x), \quad (29)$$

where φ satisfies the Poisson equation in the Solar-System regime. The background defines a preferred cosmological rest frame, but the rest frame is not fixed: it is determined dynamically by the scalar field gradient.

To leading order,

$$U_\mu \simeq (-1, 0, 0, 0), \quad (30)$$

and the weak-field metric reduces to

$$g_{00}^{(m)} = -(1 + 2\varphi), \quad g_{ij}^{(m)} = (1 - 2\varphi)\delta_{ij}, \quad (31)$$

yielding the standard PPN result $\gamma = \beta = 1$.

D.2 Boosted Frame

We now transform to a frame moving with velocity $\mathbf{w}$ relative to the cosmological rest frame. The Lorentz transformation gives

$$\partial_\mu \Phi \rightarrow \partial'_\mu \Phi = \Lambda_\mu{}^\nu \partial_\nu \Phi, \quad (32)$$

so that the normalized vector becomes

$$U'_\mu = (-1 + \mathcal{O}(w^2), w_i + \mathcal{O}(w\varphi)). \quad (33)$$

Substituting into the matter metric and expanding to post-Newtonian order:

$$g_{00}^{(m)} = -(1 + 2\varphi) + \mathcal{O}(w^2\varphi), \quad (34)$$

$$g_{0i}^{(m)} = \mathcal{O}(w\varphi), \quad (35)$$

$$g_{ij}^{(m)} = (1 - 2\varphi)\delta_{ij} + \mathcal{O}(w^2\varphi). \quad (36)$$

D.3 Extraction of α_1 and α_2

In the PPN formalism, the preferred-frame parameters are defined by the coefficients of the w_i -dependent terms in g_{0i} and g_{ij} . Comparing with the standard PPN metric, we identify

$$g_{0i}^{(m)} = -\frac{1}{2}\alpha_1 w_i U + \mathcal{O}(U^2), \quad g_{ij}^{(m)} = (1 + 2\gamma U)\delta_{ij} + \alpha_2 w_i w_j U + \dots, \quad (37)$$

where U is the Newtonian potential.

The boost-induced corrections to the metric are controlled by the ratio of the background scalar kinetic energy to the normalization scale f . Defining

$$\epsilon \equiv \frac{\dot{\Phi}_0^2}{Z_0 f^2}, \quad (38)$$

we find that the preferred-frame parameters scale as

$$\alpha_1 = \epsilon w^2, \quad \alpha_2 = \epsilon w^2, \quad (39)$$

so that

$$C_1 = C_2 = \epsilon = \frac{\dot{\Phi}_0^2}{Z_0 f^2}. \quad (40)$$

For Solar-System applications, $w \simeq 10^{-3}$, so $w^2 \sim 10^{-6}$. If the background evolution satisfies $\epsilon \lesssim 10^{-1}$, then

$$\alpha_1 \sim 10^{-7}, \quad \alpha_2 \sim 10^{-7}, \quad (41)$$

comfortably within the observational bounds $|\alpha_1| \lesssim 10^{-4}$ and $|\alpha_2| \lesssim 10^{-7}$ without any fine-tuning of parameters.

D.4 Summary

Because the substrate rest frame is defined dynamically by the scalar field gradient, preferred-frame effects arise only from the small boost of the Solar System relative to the cosmological background. The resulting PPN parameters scale as $w^2 \sim 10^{-6}$ and can be made consistent with observational bounds for natural choices of the scalar normalization scale f and the background evolution $\Phi_0(t)$.

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